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Reviewing the Code Review



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Agenda

- Why do we do code review
- Review the process
- Review the reviewer
- Review the reviewee
- Are Copilot going to help?



Why Code Review?

- Ensuring code quality
- Following coding guidelines
- Find bugs and edge cases
- Avoiding the lottery factor of 1
- Knowledge transfer



Why Code Review

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"Ask yourself "If I had to support this in production tomorrow, with no previous knowledge of it, how tempted would I be to send a glitter bomb to the developer who wrote it?"

When you think like that, it really cleans up a lot of your code"
– Unknown Redditor

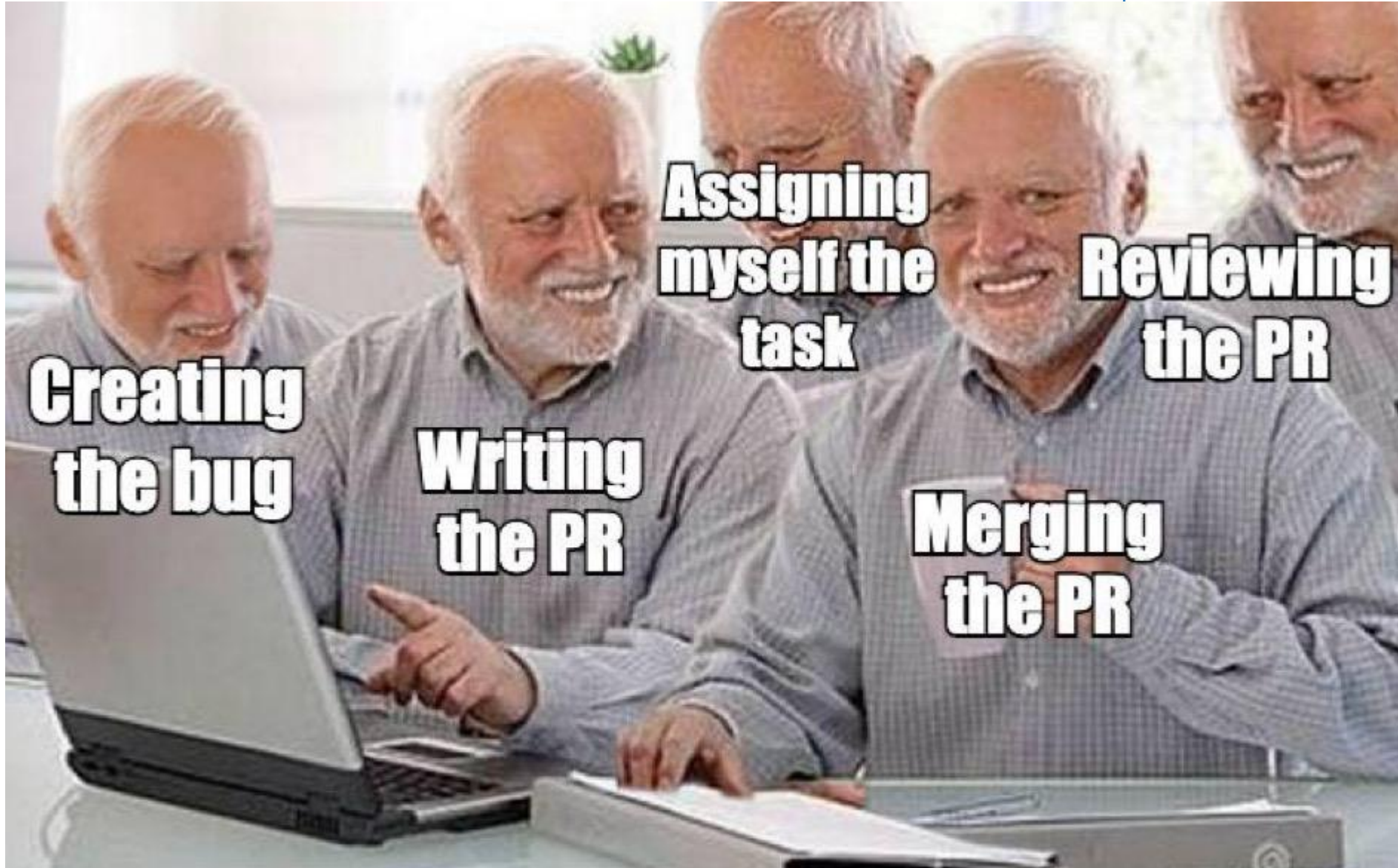
Why Code Review?

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Why Code Review?

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Review the process

Pull Requests, policies, pipelines and linters

- Does it build?
- Do all the tests pass?
- Are translations done?
- Is it following the coding guidelines?



Linting use cases – Coding Conventions

You, 54 seconds ago | 1 author (You)

var

2 references

HideMessage: Boolean;

(global) HideMessage: Boolean

Global variables should be prefixed with "g" AL(TS0002)

[View Problem \(Alt+F8\)](#) [Quick Fix... \(Ctrl+.\)](#)

```
if SkipCheck then  
    exit;
```

The exit keyword ends the procedure with the specified exit value.

EXIT keyword must be Capitalized AL(TS0005)

[View Problem \(Alt+F8\)](#) [Quick Fix... \(Ctrl+.\)](#)



Linting use cases – Maintainability improvements

1 reference | 0% Coverage

```
local procedure ImportVendorBankAccounts()  
begin  
    ImportBankAccountsFromExcel();  
  
    Commit();  
  
    AssignPrimaryBankAccount();  
end;
```

1 reference | 0% Coverage

```
local procedure CreateNewCustomer(Number: Code[20]; Name: Text[100])  
var  
    Customer: Record Customer;  
begin  
    Customer.Init();  
    Customer."No." := Number;  
    Customer.Name := Name;  
    Customer.Insert();  
end;
```

```
procedure Commit()
```

Ends the current write transaction.

Commit() needs a comment to justify its existence. Either a leading or a trailing comment. AL(TS0012)

[View Problem \(Alt+F8\)](#) [Quick Fix... \(Ctrl+.\)](#)

```
procedure Insert(): Boolean
```

Inserts a record into a table without executing the code in the OnInsert trigger.

Internal Methods must be invoked with explicit parameters AL(TS0013)

[View Problem \(Alt+F8\)](#) [Quick Fix... \(Ctrl+.\)](#)



Review the process

Pull Requests, policies, pipelines and linters

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Review the process

Size, time and frequency of pull requests

- Many smaller pull requests are better than one huge
- Make small pull requests
- No, seriously, keep pull requests small



Small work item – Small Pull Request

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“But when the code's finished is too late for deep feedback. Any pushback breeds resentment. What do you mean you don't like my beautiful baby!?”
– Swizec Teller

<https://swizec.com/>

Review the process

Size, time and frequency of pull requests

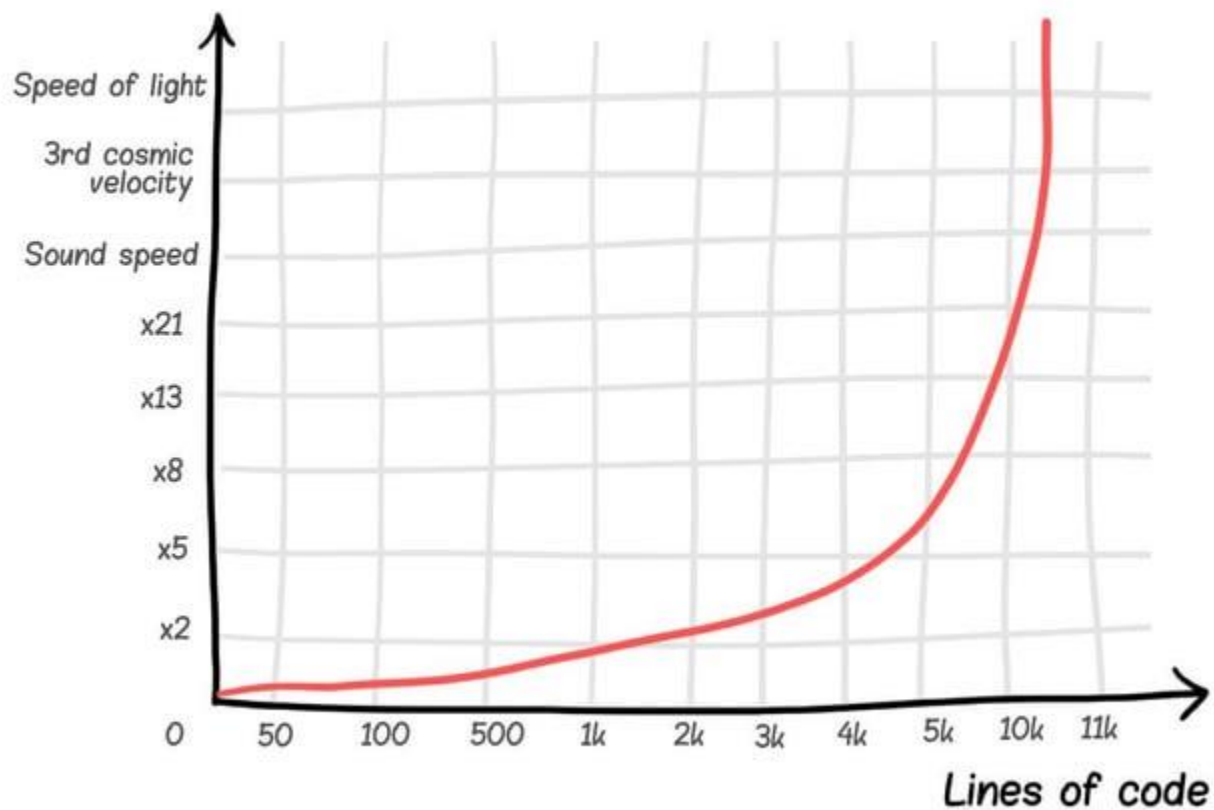
- Many smaller pull requests are better than one huge
- Make small pull requests
- No, seriously, keep pull requests small
- Small work items will result in a small PR.
- Ideal work span is up to 3 days, 5 is okay
- 1 hour is too little, 2 weeks are questionable, and more than two weeks should never happen.
- If you're doing big PRs, plan time for code review



Make it small!

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Code Review Scrolling Speed



Make it small!

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Ryan Peterman ✓
@ryanlpeterman

Ask an engineer to review a 15 lines of code and they'll find plenty of issues.

Ask them to review 1000 lines and they'll say it looks good.

44 Retweets 661 Likes

Reviewing the Reviewer

- Be timely – Prioritize review work to unblock others
- Review the changes not just the syntax
- Stick to the scope



Review

Coding guide

- Differences
- The ultimate

<https://algorithms.wtf>

Code formatting

Code should follow the AL formatting rules enforced by VS Code. No individual formatting rules should be set because different formatting ends up with unnecessary changes in each commit.

In general, there is no restriction on the line length, but lengthy lines can make the code unreadable. We recommend having a max length of 120 symbols to keep code easily scannable and readable.

When considering the issue of excessively long procedure signatures or invocations, it becomes imperative to maintain code readability and conformity to established coding standards. In cases where a procedure signature or invocation spans an excessive length, it is recommended to break the parameters into separate lines to enhance code clarity. Proper indentation plays a pivotal role in achieving this objective. An exemplary illustration of correct indentation involves aligning the parameters in a manner that facilitates quick comprehension, typically by placing each parameter on a new line and indenting them appropriately, as demonstrated below:

```
internal procedure SetAddress(  
    Address: Text[100];  
    Address2: Text[50];  
    PostCode: Code[20];  
    City: Text[30];  
    County: Text[30];  
    CountryCode: Code[10];  
    Contact: Text[100]  
)  
begin  
end;
```

```
trigger OnRun()  
begin  
    SetAddress(  
        Address,  
        Address2,  
        PostCode,  
        City,  
        County,  
        CountryCode,  
        Contact  
    )  
end;
```



Reviewing the Reviewer

Clear communication is key

Five types of code to look for:

- Good code
- Could-be-better code
- Bad code
- “Meh” code
- Amazing code

Adopt the power of “nit”



Review the Reviewee

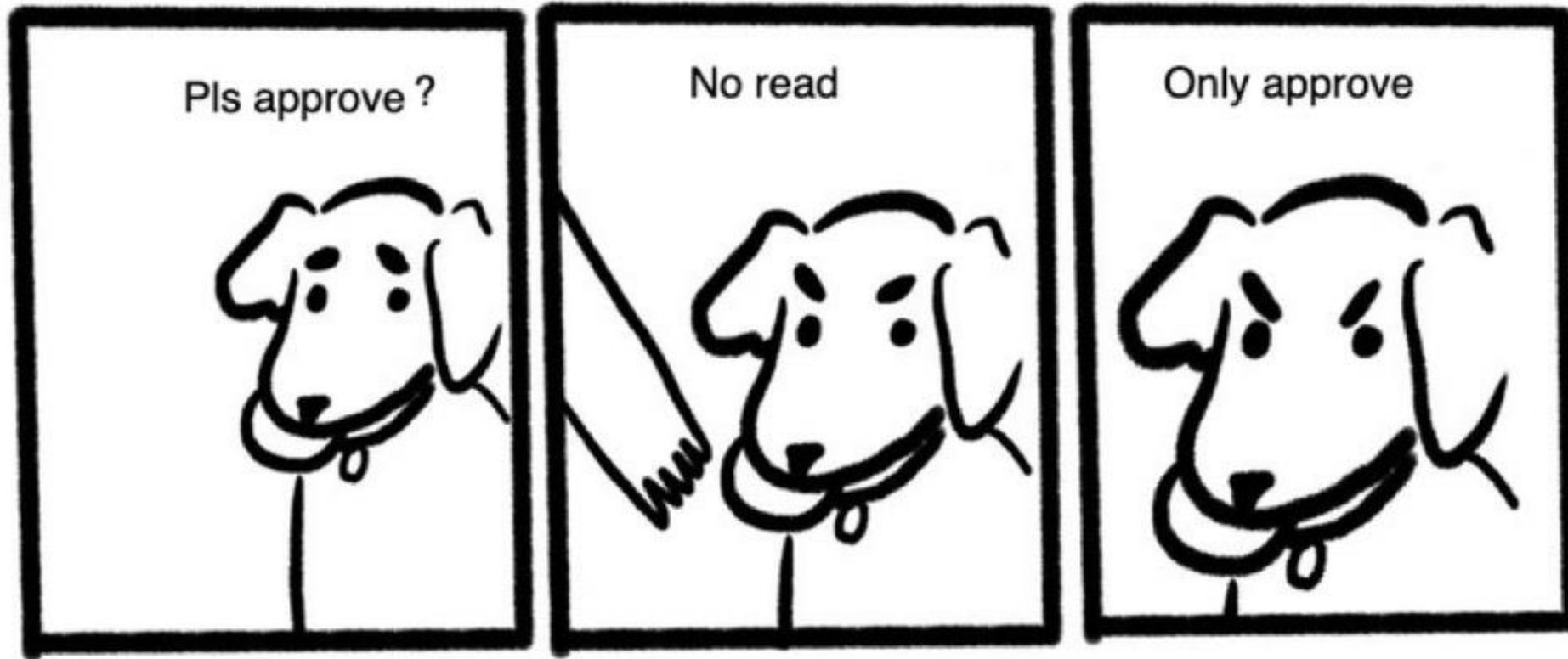
- Review yourself
- Be proactive with clarifying unclear parts
- Be open to feedback



Review the Reviewee

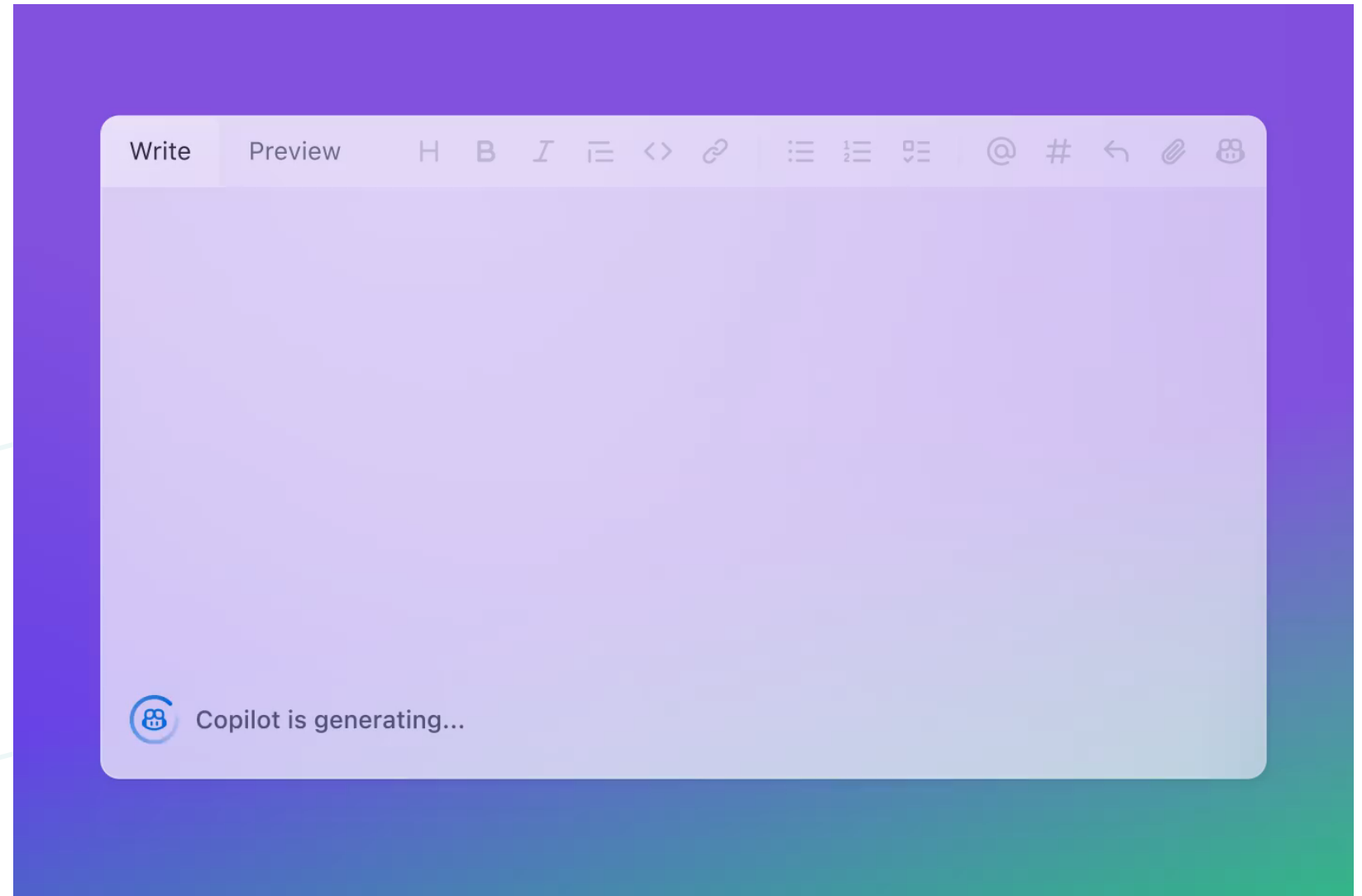
“..I need to push this PR fast..”

“..I don't have time to argue..”



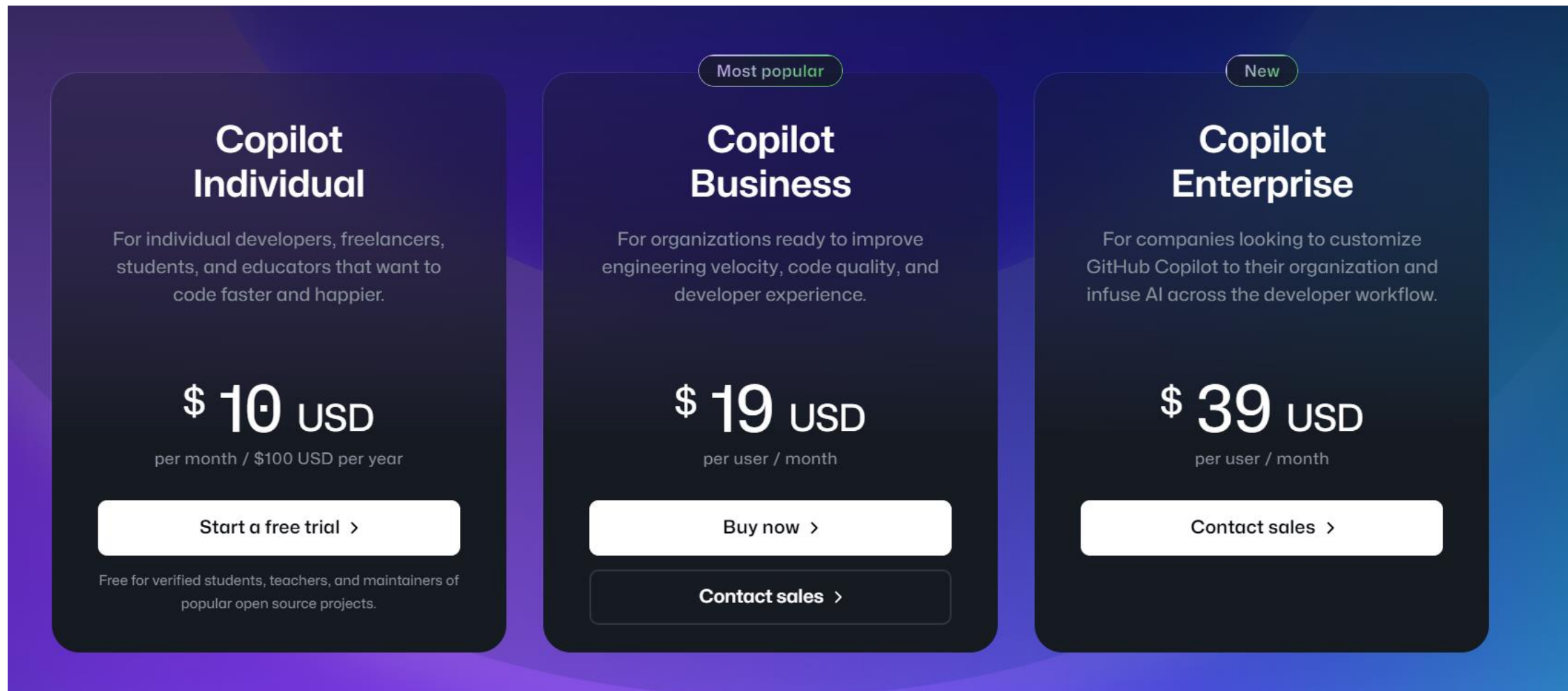
GitHub Copilot

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GitHub Copilot

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The image shows the GitHub Copilot pricing page with three main cards: Copilot Individual, Copilot Business, and Copilot Enterprise. Each card includes a description, pricing, and a call-to-action button. The Copilot Business card is marked as 'Most popular' and the Copilot Enterprise card is marked as 'New'.

Plan	Target Audience	Price	Call to Action
Copilot Individual	For individual developers, freelancers, students, and educators that want to code faster and happier.	\$10 USD per month / \$100 USD per year	Start a free trial >
Copilot Business	For organizations ready to improve engineering velocity, code quality, and developer experience.	\$19 USD per user / month	Buy now >
Copilot Enterprise	For companies looking to customize GitHub Copilot to their organization and infuse AI across the developer workflow.	\$39 USD per user / month	Contact sales >

Copilot Individual

For individual developers, freelancers, students, and educators that want to code faster and happier.

\$10 USD

per month / \$100 USD per year

Start a free trial >

Free for verified students, teachers, and maintainers of popular open source projects.

Copilot Business

For organizations ready to improve engineering velocity, code quality, and developer experience.

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Copilot Enterprise

For companies looking to customize GitHub Copilot to their organization and infuse AI across the developer workflow.

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Free

The basics for individuals
and organizations

\$0 per user/month

Create a free organization

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Advanced collaboration for
individuals and organizations

\$4 per user/month

Continue with Team ▾

RECOMMENDED

Enterprise

Security, compliance,
and flexible deployment

\$21 per user/month

Start a free trial

Contact Sales

Key Takeaways

- Do the code review
- Focus on design before code
- Keep the pull requests small
- Adopt process improvements to make reviews faster
- Clearly communicate your intention

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Small work item – Small Pull Request

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“We need to understand that PRs aren’t an obstacle that we need to overcome, but a tool to collectively have better code.”
– James Pearson

Give us Feedback!

Find me on:

- X (Twitter): @TineStaric
 - LinkedIn: /timestaric
 - Blog: tine.staric.net
 - Email: tine.staric@companial.com
-
- Reach out with questions!



Thank you!

